
Course Name: Machine Learning for Earth System Sciences

Week 06 - Assignment 06 - (Jul-2026)

TYPE OF QUESTIONS: MCQ/MSQ

Number of questions: 10

Total marks: 10x1= 10

QUESTION 1

The interpretation of neural networks can enable the discovery of scientifically meaningful connections within geoscientific data. In accordance to the neural network interpretations identify the False statements from the below options.

- (A) Backward optimization is a neural network interpretation technique and is also known as optimal input technique.
- (B) The backward optimization method takes the reverse approach of a neural network training and it always tries to update the weight, bias and input of the network to get the optimal input.
- (C) Like Backward optimization, Layer-wise Relevance Propagation is also a neural network interpretations technique.
- (D) Both Backward optimization and LRP method works with pre trained neural networks whose weights and biases are frozen after training.

Answer: B

Detailed Solution:

Backward optimization tries to update the sample input of the model to reduce the loss, but it does not update the weight and bias of the pre-trained neural network.

QUESTION 2

ConvNet is a CNN based architecture used by Gagne et.al in his paper "Interpretable Deep Learning for Spatial Analysis of Severe Hailstorms". In this study, the ConvNet model is compared with some standard machine learning approaches for predicting the probability of severe hailstorms. Which of the following machine learning models are used in this comparison?

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- (A) Logistic Mean and SVM
- (B) Logistic PCA and SVM
- (C) Logistic Mean and Logistic PCA
- (D) SVM and PCA

Answer: C

Detailed Solution:

In this study the ConvNet model is compared with the Logistic Mean and Logistic PCA for predicting the probability of severe hailstorms.

Ref: Gagne II DJ, Haupt SE, Nychka DW, Thompson G. Interpretable deep learning for spatial analysis of severe hailstorms. Monthly Weather Review. 2019 Aug;147(8):2827-45.

QUESTION 3

Which of the following is not a limitation of YOLO?

- (A) YOLO struggles with detection of small objects that appear in groups, such as flocks of birds.
- (B) YOLO struggles to generalize to objects in new or unusual aspect ratios or configuration.
- (C) YOLO is less likely to predict false positives on background.
- (D) YOLO gives incorrect localization.

Answer: C

Detailed Solution:

Option (C) is not a limitation of YOLO.

Ref: Redmon J, Divvala S, Girshick R, Farhadi A. You only look once: Unified, real-time object detection. In Proceedings of the IEEE conference on computer vision and pattern recognition 2016 (pp. 779-788).

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QUESTION 4

When the images are captured at many wavelengths within a single spectral band then it is called as:

- (A) Multispectral Image
- (B) Hyperspectral Image
- (C) Infra-red Image
- (D) Visible Image

Answer: B

Detailed Solution:

When the images are captured at many wavelengths within a single spectral band then it is called as Hyperspectral Image.

QUESTION 5

Deep Learning architectures can be classified as two-stage, single-shot, and anchor-free architecture. Identify the perfect match between the architectures and models given below

Architecture	Model
(i) Two-Stage	(a) YOLO
(ii) Single-Shot	(b) R-CNN
(iii) Anchor-Free	(c) Corner-Net

- (A) (i)-(a), (ii)-(b), (iii)-(c)
- (B) (i)-(c), (ii)-(b), (iii)-(a)
- (C) (i)-(b), (ii)-(a), (iii)-(c)
- (D) (i)-(a), (ii)-(c), (iii)-(b)

Answer: C

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Detailed Solution:

Architecture	Model
Two-Stage	R-CNN
Single-Shot	YOLO
Anchor-Free	Corner-Net

Ref: Sharma M, Dhanaraj M, Karnam S, Chachlakis DG, Ptucha R, Markopoulos PP, Saber E. YOLOrs: Object detection in multimodal remote sensing imagery. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing. 2020 Nov 30;14:1497-508.

QUESTION 6

“DOTA: A Large-scale Dataset for Object Detection in Aerial Images” is a research article by Xia et. al. To show the usefulness of DOTA in horizontal object detection the authors have evaluated some state-of-the-art object detection algorithms on DOTA. Which of the following algorithms are used in this task?

- (i) Faster R-CNN (ii) R-FCN (iii) SSD (iv) YOLOv2
- (A) (i) and (ii)
(B) (i), (ii) and (iii)
(C) (i), (ii), (iii) and (iv)
(D) (ii), (iii) and (iv)

Answer: C

Detailed Solution:

For horizontal object detection with DOTA, the authors have used Faster R-CNN, R-FCN, YOLOv2 and SSD as the benchmark testing algorithms.

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Ref: Xia GS, Bai X, Ding J, Zhu Z, Belongie S, Luo J, Datcu M, Pelillo M, Zhang L. DOTA: A large-scale dataset for object detection in aerial images. In Proceedings of the IEEE conference on computer vision and pattern recognition 2018 (pp. 3974-3983).

QUESTION 7

The CNN based method proposed by Fu et.al for Hyperspectral Image Super-Resolution (Ref: "Hyperspectral Image Super-Resolution with Optimized RGB Guidance", Fu et.al. 2019) is

- (A) A supervised method
- (B) An unsupervised method
- (C) A semi-supervised method
- (D) Either supervised or semi-supervised method.

Answer: B

Detailed Solution:

The authors have presented an unsupervised CNN-based method for HSI super-resolution.

Ref: Fu Y, Zhang T, Zheng Y, Zhang D, Huang H. Hyperspectral image super-resolution with optimized RGB guidance. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition 2019 (pp. 11661-11670).

QUESTION 8

Which of the following statement is true for a panchromatic (PAN) image?

- (A) PAN images possess high spatial resolution but low spectral resolution.
- (B) PAN images possess low spatial resolution and low spectral resolution.
- (C) PAN images possess high spatial resolution and high spectral resolution.
- (D) PAN images possess low spatial resolution but high spectral resolution.

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Answer: A

Detailed Solution:

PAN images possess high spatial resolution but low spectral resolution.

Ref: Shao Z, Cai J. Remote sensing image fusion with deep convolutional neural network. IEEE journal of selected topics in applied earth observations and remote sensing. 2018 Mar 12;11(5):1656-69.

QUESTION 9

“SkyScapes – Fine-Grained Semantic Understanding of Aerial Scenes” is an article by Azimi et.al.(2019). From the given options identify the most appropriate option that indicates the main contribution of this work.

- (A)Preparation of an aerial image dataset with highly-accurate, fine-grained annotations for pixel-level semantic labeling.
- (B)Development of a fully connected neural network for fine-grained semantic understanding of aerial scenes.
- (C) Introduction of an image restoration technique for satellite images.
- (D) Development of a nested image segmentation network.

Answer: A

Detailed Solution:

Ref: Azimi SM, Henry C, Sommer L, Schumann A, Vig E. SkyScapes – Fine-Grained Semantic Understanding of Aerial Scenes. In Proceedings of the IEEE/CVF International Conference on Computer Vision 2019 (pp. 7393-7403).

QUESTION 10

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RDU_Net is a novel architecture used for Sea_Land Segmentation using remote sensing images. It is also assumed that there are some other architectures that can be employed for the same task. Which of the following networks can be used to perform the same task as RDU_Net.

- (A) Only FusionNet
- (B) Only DeepUNet
- (C) Both FusionNet and DeepUNet
- (D) None of these

Answer: C

Detailed Solution:

Ref: Shamsolmoali P, Zareapoor M, Wang R, Zhou H, Yang J. A novel deep structure U-Net for sea-land segmentation in remote sensing images. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing. 2019 Aug 6;12(9):3219-32.